

# Sathwik Goundla

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## PROFESSIONAL SUMMARY

AI and Machine Learning enthusiast with hands-on experience in building practical AI solutions. Skilled in Python, machine learning algorithms, and data visualization. Experience developing Generative AI systems using LangChain and RAG pipelines, predictive ML models with XGBoost and Random Forest, and interactive data visualization platforms. Strong foundation in deep learning, NLP, and translating real-world problems into data-driven solutions.

## TECHNICAL SKILLS

**Programming Languages:** Python (NumPy, Pandas, Scikit-learn, XGBoost)  
**Machine Learning:** Random Forest, XGBoost, SMOTE, Feature Engineering, Model Evaluation  
**Generative AI:** LangChain, FAISS, RAG Pipelines, LLMs, Embeddings  
**Data Visualization:** Streamlit, Plotly, Folium, Matplotlib  
**Model Explainability:** SHAP, LIME  
**Tools & Libraries:** Git, Jupyter, VS Code

## AI/ML PROJECTS & EXPERIENCE

- Retrieval-Augmented Generation (RAG) System** | *Python, LangChain, FAISS, LLMs* 2026
  - Architected production-grade RAG pipeline integrating LangChain, FAISS vector database, and LLMs for context-aware document retrieval and response generation achieving 90%+ accuracy with 55% latency reduction
  - Engineered end-to-end AI workflow with semantic search using embeddings, advanced prompt engineering, and retrieval optimization delivering real-time inference capabilities
  - Built scalable Python application with modular architecture supporting 100+ concurrent queries and seamless integration
- Predictive ML Loan Default Classification System** | *Python, XGBoost, Random Forest, SHAP* 2025
  - Developed production-ready loan default prediction pipeline using XGBoost and Random Forest achieving 88% accuracy with 0.936 AUC on 1,000+ records, supporting credit risk assessment with SMOTE balancing improving minority class prediction by 23%
  - Integrated SHAP and LIME explainability frameworks providing transparent model insights for regulatory compliance and stakeholder decision-making in financial systems
  - Built modular scikit-learn architecture with automated ETL pipeline, cross-validation, and comprehensive evaluation metrics ready for production deployment
- StarGuard: Asteroid Impact Risk Visualization Platform** | *Python, Streamlit, Plotly, Folium* 2024
  - Engineered interactive web application for real-time asteroid impact risk assessment featuring global geospatial maps, animated trajectory visualizations, and physics-based crater simulator using Streamlit, Folium, and Plotly
  - Implemented sophisticated geospatial visualization algorithms processing NASA datasets with Pandas for data-driven planetary defense awareness and actionable insights
  - Created scalable web architecture with responsive UI/UX supporting real-time data updates and interactive scenario modeling

## EDUCATION

- Bachelor of Technology in Computer Science Engineering** August 2023 – April 2027  
*Sphoorthy Engineering College* *Hyderabad, Telangana, India*
  - Relevant Coursework:** Machine Learning, Deep Learning, NLP, Data Structures, Algorithms, DBMS, Cloud Computing

## CERTIFICATIONS & PROFESSIONAL DEVELOPMENT

Deep Learning Specialization (Coursera) | Advanced NLP with Transformers | Generative AI with LLMs (DeepLearning.AI) | LangChain for LLM Development | AWS ML Foundations